The Momentum Mystery: Demystifying the mystery

Summary

Tennis more than any other sport, is a game of momentum. The absence of a clock to do the dirty work of finishing off an opponent, and a scoring system based on units used, makes the flow of the match much more important than any lead that has been established."

For Program I: After outliers were processed for player1_score and player2_score, a data visualization was performed for the cumulative points of the 2023 Wimbledon men's singles final match between Djokovic and Alkaraz. Based on this match and other top 32 matches of men's singles in Wimbledon this year, the **TOPSIS entropy weight** method is established, which **takes serving, stroking and winning as the first-level index and ace ball, double fault and run distance as the second-level index**. It is obtained that ace, net and break account for the largest proportion in serving, stroking and winning situations respectively. Then the three first-level indicators are assigned. The formula of momentum for three primary indicators and a total of four formulas for three primary indicators corresponding to two indicators are obtained. Finally, the formula is applied to the men's singles final mentioned before, and the momentum of each set is further visualized.

For Program II: A **Spearman-based correlation analysis** was established for four matches on the way to Sinna and five matches on the way to Djokovic at Wimbledon. The point at which momentum turns from decline to rise is the upward turning point (trough), and the point at which momentum turns from rise to fall is the downward turning point (peak). Spearman's correlation analysis takes **the difference between the number of upward turning points and downward turning points** in a match as the independent variable, and the outcome of each match as the dependent variable. The correlation between the two is **0.756**, which is relatively significant.

For Program III: Initially, the logistic regression model is used to further analyze momentum performance fluctuations. Based on the secondary indicators, after some modifications, the indicators that have an impact on momentum real-time performance fluctuations are taken as independent variables, and momentum fluctuations are taken as dependent variables for **Logistic regression analysis**. The corresponding formula is obtained, and the conclusion is drawn that all are significant except for double faults, score changes and break. Finally, the paper analyzes the **CRITIC** weight of each dependent variable and finds that the break and ace are the most important factors in the influence of momentum fluctuation. We have found the relationship between real-time fluctuations in momentum and several indicators, so we suggest that when a player plays a new match with another player, it should be compared to the opponent, **intensive training, adjustment of game strategy, and observation of the opponent's forehand and main hand**.

For Program IV： In this paper, the logistic regression analysis model is applied to Medvedev's four matches to test the logistic regression model. It is found that the agreement **between the model prediction results, and the actual competition data is 70.99%**. To make the results obtained by the model more accurate, we introduced a new variable, **"ATP World Ranking of men's Tennis Singles"**, into the future model. This model has strong universality to women's tennis matches and various tennis tournaments, as well as various tennis courts such as clay courts, hard courts, and grass courts, but it has poor universality to other ball games such as table tennis due to large differences in rules and scoring methods.

Keywords: TOPSIS, Spearman, Logistic regression model. CRITIC weight method

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1 Introduction

1.1 Problem Background

In sports, particularly in individual games like tennis, the concept of "momentum" is frequently discussed yet challenging to quantify. Observations of match dynamics, such as a player's comeback or sudden loss of advantage, suggest momentum's potential influence on the outcome, yet these perceptions are subjective and prone to bias. The thrilling match between Carlos Alcaraz and Novak Djokovic in the 2023 Wimbledon final exemplifies the complexity of match dynamics and the potential role of momentum.

To objectively assess and quantify momentum, a model that captures the fluid nature of a match is required. This model must account for unique aspects of tennis, like the server's higher probability of winning points and games and differentiate between performance fluctuations driven by momentum and those resulting from random variation. By applying such a model, not only can the scoring be analyzed, but also the context behind each point, offering a comprehensive understanding of the match flow, player performance, and the actual impact of momentum in the game. This analysis is valuable for fans, commentators, coaches, and players alike, as understanding and leveraging momentum in matches can be crucial for performance improvement and strategic planning.



Figure 1 The Championships, Wimbledon

1.2 Restatement of the problem

The 2023 Wimbledon men's final showcased an intense battle where 20-year-old Spanish sensation Carlos Alcaraz overcame the 36-year-old Novak Djokovic, marking a rare defeat for Djokovic at Wimbledon. This match, filled with remarkable shifts in momentum, prompts an in-depth analysis of such fluctuations in tennis. To understand and quantify momentum shifts in matches like these, we need to address several points:

- Construct a model that captures the flow of the match, accurately reflecting shifts in players performances over time.
- Implement this model in the Alcaraz vs. Djokovic match, providing a visual representation of the momentum shifts throughout the game.
- Integrate the server's advantage into the model, recognizing its significant impact on the game's dynamics.

- Examine the provided dataset, particularly focusing on Djokovic's strong start in the first set, to understand the scoring dynamics in detail.
- Explore the concept of momentum in sports, determining whether performance fluctuations are a result of this phenomenon or random variance, using the model to analyze match data.

1.3 Our work && model overview

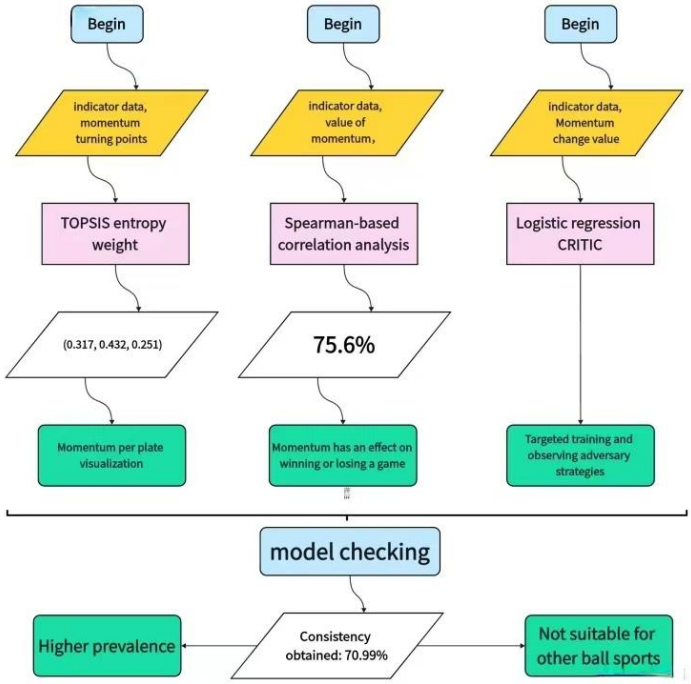


Figure 2 Our work

2 Assumptions

Since we can only use the data provided, we make the following assumption:

- Inherent Momentum in Tennis: Accept that tennis, more than any other sport, inherently exhibits momentum shifts due to its unique scoring system and the absence of a time limit to conclude the match. This assumption acknowledges the significant role of momentum in the flow of a tennis match.
- Reliability of Data and Indicators: Assume that the data collected, including scores, serving statistics, hitting patterns, and player movement (run distance), accurately reflect the players' performance and the match dynamics. This also involves trusting the TOPSIS entropy weight method and the CRITIC weight method to effectively weigh and integrate these indicators.
- Validity of Selected Indicators: Presume that the primary and secondary indicators chosen (serving, hitting, winning, ace ball, double fault, run distance, etc.) are significant contributors to a player's momentum and can be quantified and analyzed to reflect the match's flow.
- Consistency in Indicator Relevance: Assume that the proportion of significance of indicators like ace, net, and break remains consistent across different matches and can be generalized to analyze momentum.

・Correlation Reflects Causation: In the Spearman-based correlation analysis, assume that the relationship between the number of turning points (troughs and peaks) in a match and the match outcome is not just correlated but also has a causal relationship, reflecting the impact of momentum shifts on the match result.

・Logistic Regression Model Applicability: Assume that the logistic regression model and the variables selected (excluding double faults, score changes, and break) reliably predict momentum fluctuations and that these predictions are significant and applicable to different matches and players.

・Generalizability of Models: Presume that the models developed are not only applicable to the specific matches and players they were derived from but can also be generalized to other matches, including women's tennis, and various surface

3 Notations

Some important naming of indicators used in this paper are listed in Table 1.

Table 1 Indicators naming rule

Naming	Opinion
spread	difference between two players' data
sum	sum of two players' data
determine	manual assignment
score	difference between two players' data
artificial assignment	sum of first-level indicators or moment
change	impact of indicators on changes in momentum

Table 2 Notations

Symbol	Description
e_j	entropy of item J
d_j	information entropy redundancy (variance)
w_j	weights of the indicators
s_i	composite score for each sample
E_i	entropy of the i index
$\emptyset$	a vector of the obtained weights
FQ	Serve situation
JQ	Stroke situation
YQ	Win situation
M	momentum

4 Model Preparation

4.1 Indicators Determination

Among the fundamental phases of a tennis player's performance, serving, hitting and scoring are crucial aspects of the game. In the serve phase, the player launches the ball by controlling his momentum to ensure accurate ball speed and direction. When hitting the ball, the player adjusts his momentum to respond to the incoming ball from the opponent and prepares for the counterattack. The scoring phase is the peak of momentum management, in which the player regulates the transfer and distribution of momentum in an effort to dominate the exchange of balls and ultimately win the score with a precise and powerful strike. So we use these three aspects as primary indicators, and we combine them with the variables column in the file data_dictionary as secondary indicators to build our model. Since CTL, NCTL, NA, ND,D are fixed class data, we need to manually assign them to be converted to quantitative data, where CTL and D are assigned to 1, NCTL and ND are assigned to 0, and NA is assigned to -1.

Table 3: Indicators Selected in TOPSIS

Object	Indicators	Description	Positive/Negative
Serve situation(FQ)	SO	serve_no	Negative
	SA	sum_ace	Positive
	SD	sum_double_fault	Negative
	SM	speed_mph	Positive
	SP	serve_depth_determine	Positive
Stroke situration(JQ)	WS	winner_spread	Positive
	SU	sum_unf_error	Negative
	SN	sum_net_pt	Positive
	DR	distance_run_spread	Positive
	RD	return_depth_determine	Positive
Win situation(YQ)	SS	sets_spread	Positive
	GS	games_spread	Positive
	SC	scores_spread	Positive
	BD	break_pt_determine	Positive

According to the general tennis rules and match conditions, the assigned indicators are shown in the figure 3 below The break_pt_determine is initially set to 2 when a break point occurs, and if the break point is successfully honored, 1 is added to 2 to make 3. If the break point is not successfully honored, 1 is subtracted from 2 to make 1.

Before we apply the model, we first assign specific values to each indicator, as shown in the figure 4, 5, 6. True and false in the figure below refer to whether it is server or not, and first and second refer to first or second shots.



Figure 3 Assigning rules for winning situations



Figure 4 Serve situation's scoring rule

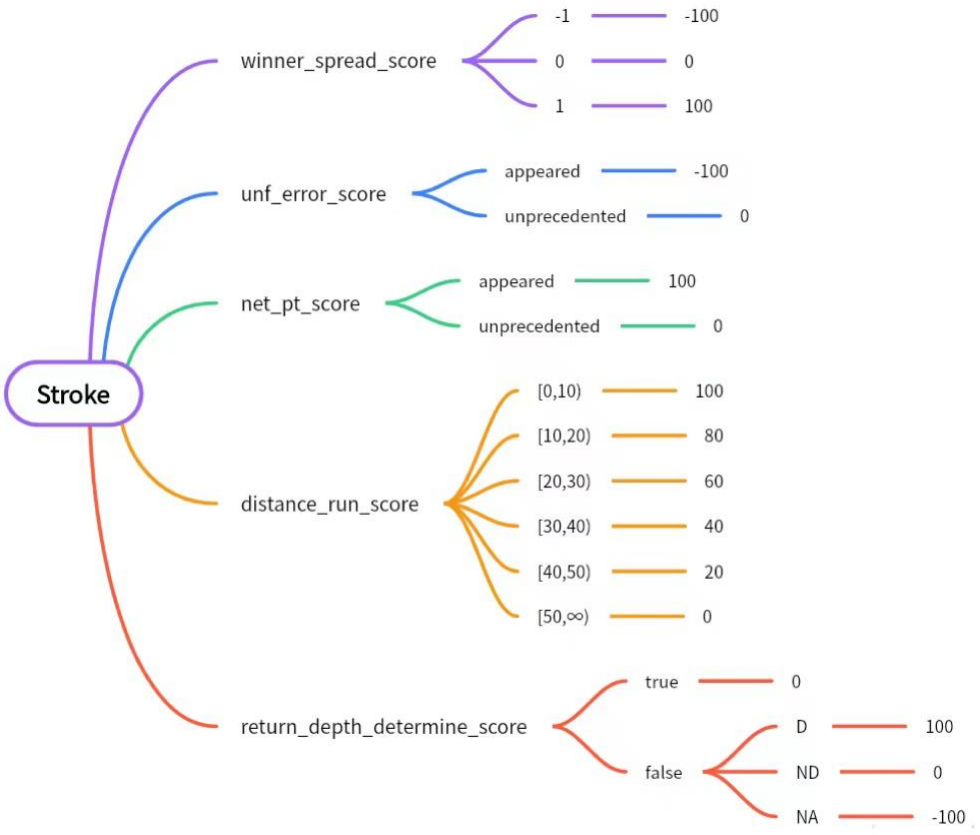


Figure 5 Stroke situation's scoring rule

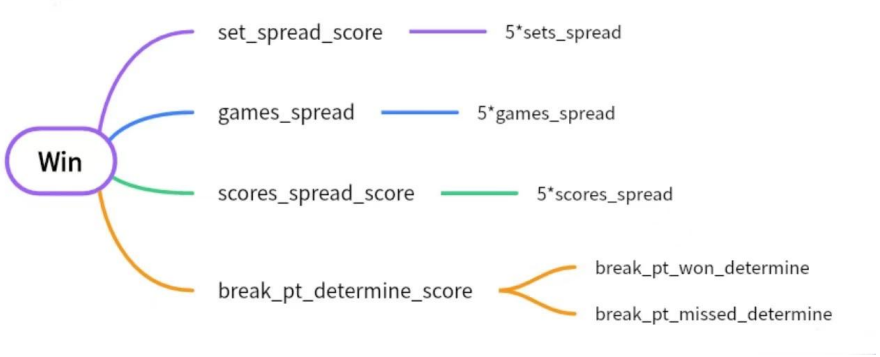


Figure 6 Win situation's scoring rule

4.2 Topsis entropy weight method analysis

Since there are "positive indicators" and "negative indicators" in Table3, these two types of data will be pre-processed separately. Here on the minimum value minus 0.0001, on the maximum value plus 0.000 is in order to be compatible with a whole column are the same value of the case, the overall results of the impact is not significant, can be ignored

$$X_{min} = min\,(X_{1j}, X_{2j}, \cdots, X_{nj}) - 0.0001 \quad (1)$$

$$X_{max} = max\,(X_{1j}, X_{2j}, \cdots, X_{nj}) + 0.0001 \quad (2)$$

For positive indicators:

$$z_{ij} = \frac{X_{ij} - X_{min}}{X_{max} - X_{min}} \tag{3}$$

For negative indicators:

$$z_{ij} = \frac{X_{max} - X_{ij}}{X_{max} - X_{min}} \tag{4}$$

Calculate the entropy of item J:

$$e_j = -k \sum_{i=1}^{n} p_{ij} \ln(p_{ij}), j = 1, \cdots, m \tag{5}$$

Calculate information entropy redundancy (variance):

$$d_j = 1 - e_j, j = 1, \cdots, m \tag{6}$$

Calculation of the weights of the indicators:

$$w_j = \frac{d_j}{\sum_{j-1}^{m} d_j}, j = 1, \cdots, m \tag{7}$$

Calculate the composite score for each sample:

$$s_i = \sum_{j=1}^{m} w_j x_{ij}, i = 1, \cdots, n \tag{8}$$

Where x_{ij} is the normalized data, based on the score of each influencing factor, the importance ranking of all factors can be obtained

The weighting analysis calculations based on the tosopsis entropy weighting method yielded the following results:

Table 4 TOPSIS analysis of serve situation

	Serve situation	
Term	information entropy e	weights(%)
sum-ace	0.734	58.557
speed_mph	0.994	1.406
serve_depth	0.865	29.666
determine_sum	0.953	10.371
double-faults	1.000	0.000

Table 5 TOPSIS analysis of stroke situation

	Stroke situation	
Term	information entropy e	weights(%)
winner_spread	0.877	28.913

speed_mph	0.837	38.410
serve_depth	0.924	17.850
determine_sum	0.971	6.804
double_faults	0.966	8.024

Table 6 TOPSIS analysis of win situation

	Win situation	
Term	information entropy e	weights(%)
sets_spread	0.928	14.206
games_mph	0.945	10.849
scores_depth	0.932	13.403
break_pt_determine	0.687	61.543

In order to visualize the weight ratios more, we created percentage doughnut charts:

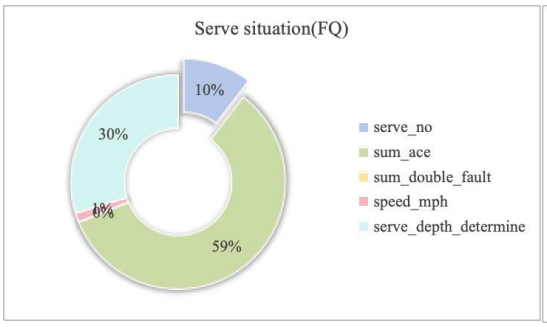


Figure 7(a) serve situation percentage circle chart

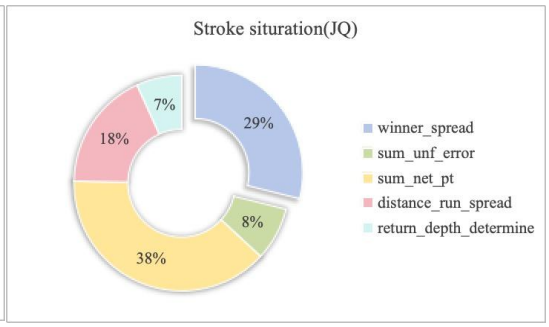


Figure 7(b) stroke situation percentage circle chart

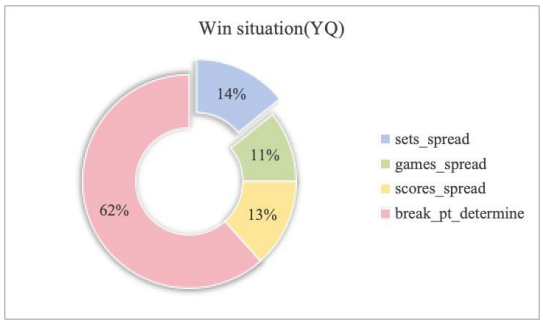


Figure7(c) win situation percentage circle chart

Considering that 2023 Men's Wimbledon is one of the top tennis tournaments in the world, both players of the tournament have great strength in tennis. At the same time, the number of their double-fault is 0 and the break-pt-determine will be far greater than any other race. However, this situation is a variable that cannot be ignored and can have a significant impact on momentum in an actual match. Therefore, the data other than double_fault used in the entropy weighting method were multiplied by 80% and break-pt-determine by 60%, and the rest of these two panels were adjusted by the corresponding proportions. The following figure shows the adjusted weights:

Table 7 Weight of the secondary indicator

Object	Indicators	Weight
Serve situation(FQ)	SO	0.0832
	SA	0.468
	SD	0.2
	SM	0.0112
	SP	0.2376
Stroke situration(JQ)	WS	0.28913
	SU	0.08024
	SN	0.3841
	DR	0.1785
	RD	0.06804
Win situation(YQ)	SS	0.22335
	GS	0.16988
	SC	0.21077
	BD	0.396

5 Model Preparation

Before developing the model, visualize the scores of Alcalas and Djokovic:

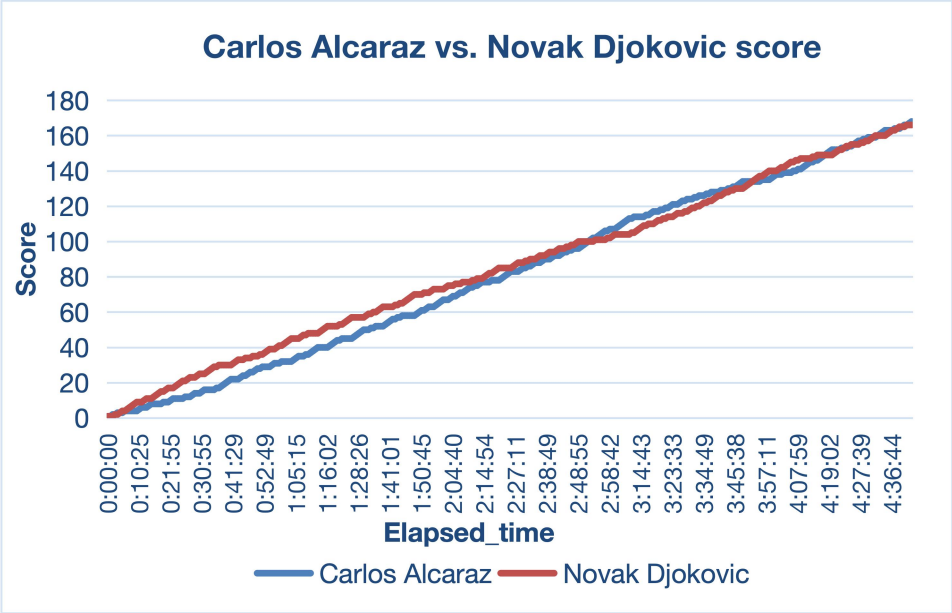


Figure 8 Carlos Alcaraz vs. Novak Djokovic score

By observing the score trend chart, it is found that in the beginning of the match, Djokovic was in the advantage, but in the middle of the match, Alkaras gradually adjusted back to the state and occupied the initiative, although in the back end of the match, Djokovic regained the form for a period of time, but finally lost to Alkaras.

5.1 Model building

Then, calculate the entropy E_i of the i index:

$$E_i = -\frac{\sum_{j=1}^{m} \tilde{p}_{ij} \cdot \ln p_{ij}}{\ln m} \left(i = 1, 2, \cdots, Num_{serve}/Num_{stroke}/Num_{Win} \right) \quad (9)$$

Based on the information entropy, we will further calculate the weight w of each evaluation indicator we defined before is calculated as follows:

$$\omega_i = \frac{1-E_i}{\sum_{i=1}^{n}(1-E_i)} \tag{10}$$

We form a vector of the obtained weights and call the vector as "evaluation vector"denoted as $\vec{\phi}$ = (ϕ , ϕ_2, ... , Num_{Serve} / Num_{Stroke} / Num_{Win}); By composing a vector of the obtained influences of different aspects and calling this vector as "influence degree vector", denoted as $\vec{S}$ = (S_1, S_2, ... , S_{serve} /S_{stroke}/S_{win}), then the final evaluation value is:

$$Score_x = \vec{\phi} \bullet \vec{S^T} = \sum_{i=1}^{Num_x} \phi_i \bullet S_i \tag{11}$$

where x represents "Serve situation", "Stroke situation" or "Win situation".

5.2 Model solving

Based on the above theory, we establish the following formula:

$$\sum M = \sigma_1 \times FQ + \sigma_2 \times JQ + \sigma_3 \times YQ \tag{12}$$

$$\sum FQ = \alpha_1 \times SO + \alpha_2 \times SA + \alpha_3 \times SD + \alpha_4 \times SM + \alpha_5 \times SP \tag{13}$$

$$\sum JQ = \beta_1 \times WS + \beta_2 \times SU + \beta_3 \times SN + \beta_4 \times DR + \alpha_4 \times RD \tag{14}$$

$$\sum YQ = \gamma_1 \times SS + \gamma_2 \times GS + \gamma_3 \times SC + \gamma_4 \times BD \tag{15}$$

In order to assign weights to these three level 1 indicators and derive the final momentum assessment value, we used the hierarchical analysis method (AHP) to construct a judgment matrix to derive the weights of the three level 1 indicators.

$$\sigma = (0.317, 0.432, 0.251) \tag{16}$$

We processed the data from the Wimbledon_featured_matches_proceased table against the assignment table (tablex) as well as the four formulas (1234) to produce the output data located in the CQ, CR columns of the Wimbledon_featured_matches_proceased file. The final visualization of the real-time momentum data for the five sets of match momentum between Carlos Alcaraz and Novak Djokovic resulted in the following figure:

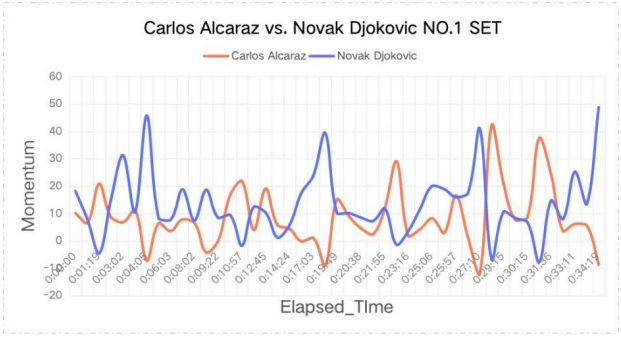


Figure9(a) First Set Real Time Momentum

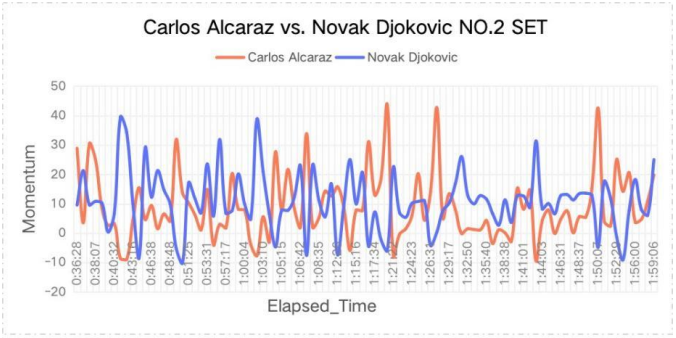


Figure9(b) Second Set Real Time Momentum

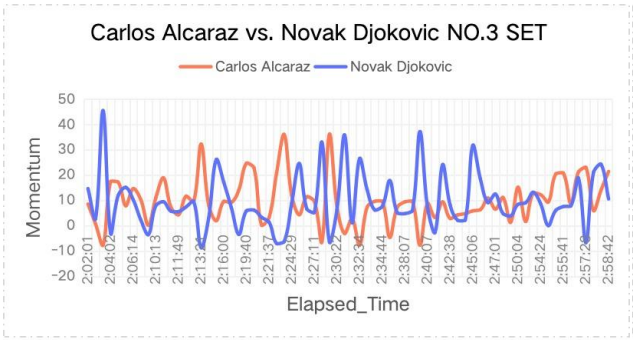


Figure10(a) First Set Real Time Momentum

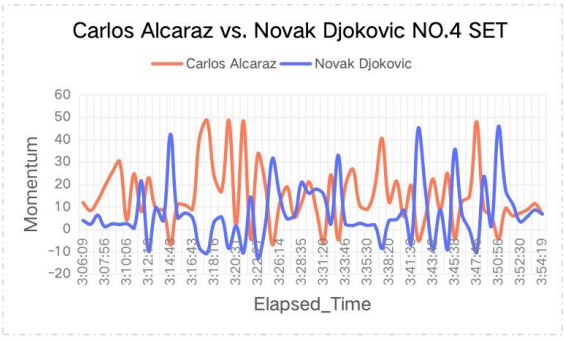


Figure10(b) First Set Real Time Momentum

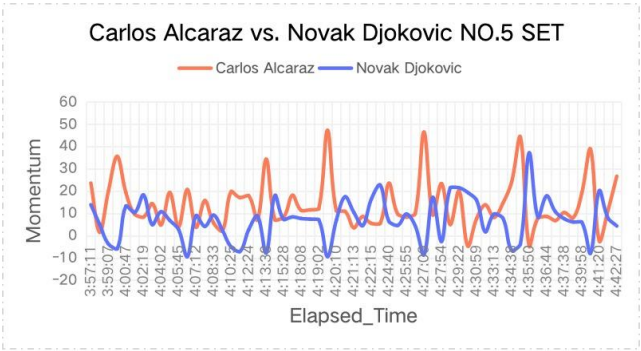


Figure11 First Set Real Time Momentum

The combined momentum graphs from the Wimbledon final show a fierce contest with frequent momentum shifts between Alcaraz and Djokovic. Alcaraz's rising momentum in later sets and Djokovic's initial dominance display a match of tactical turns and psychological edges, culminating in Alcaraz's decisive push in the final set to secure the victory. These data illustrate that momentum in tennis is inextricably linked to match situations and winning and losing.

6 Spearman model

6.1 model building

We selected the momentum data from Jannik Sinner's four and Novak Djokovic's five matches to analyze their momentum change data, i.e., the number of crests and troughs. The number of troughs and the number of peaks were made and the difference (D) was used as the independent variable and the win/loss as the dependent variable for the Spearman correlation analysis, which was carried out in the following procedure:

The Spearman correlation coefficient is defined as the Pearson correlation coefficient between rank variables. For a sample with sample size n, several raw data are converted to rank data with the following correlation coefficient p:

$$p = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - x)^2}\sqrt{\sum_i (y_i - y)^2}} \tag{17}$$

The raw data was assigned a corresponding rank based on its average descending position in the overall data. This is shown in the table below:

Table 8 Spearman hierarchy chart

Variable Xi	Descending position	Grade xi
0.8	5	5
1.2	5	5
1.2	5	5
2.3	5	5
18	5	5

In practice, the links between variables are irrelevant, so p can be calculated in simple steps. The difference between the grades of the two variables being observed, then ρ is

$$p = 1 - \frac{6\sum di^2}{n(n^2 - 1)} \tag{18}$$

6.2 Model solving

Sinner and Djokovic's relevant match data are selected for data processing to obtain the number of troughs minus peaks and the outcome of the match, Spearman model was used to analyze the relationship between the two variables and the corresponding heat maps were obtained:

Table 9 Spearman significance analysis chart

	Amount_momentum_performance_fluctuation	Win-Los
Variable x 0.8	1 (0.000***)	0.756(0.018**)
	0.756 (0.018**)	1 (0.000***)

Note: ***, **, * represent 1%, 5%, and 10% significance levels, respectively.

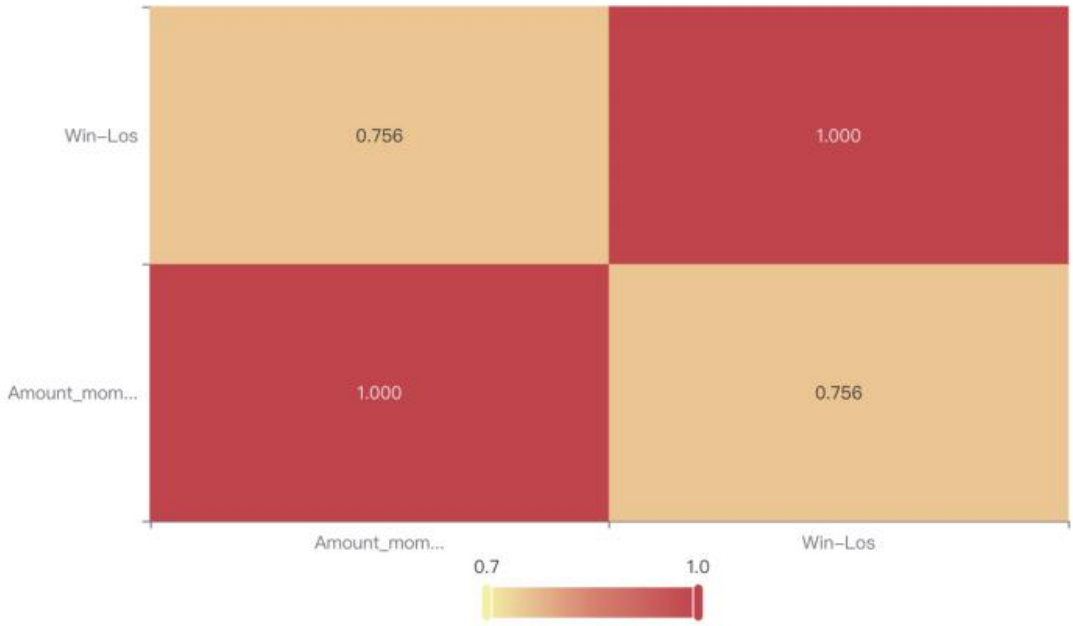


Figure 12

After the Spearman analysis above, we can get the correlation coefficient between Amount_momentum_performance_fluctuation and Win-Los is 0.756. This data shows that there is a high correlation between the two, that is, the fluctuation of a player's performance and the continuous occurrence of success should not be random. To a large extent, the fluctuation of the player's momentum performance and the continuous occurrence of success are closely related to the player's performance of serving, hitting and winning in the course of the match.

7 Constructing Momentum Models

7.1 Logistic regression

In order to develop a model that can predict the change of momentum in the competition, we take the change of each indicator as an independent variable, and take the real-time performance fluctuation of momentum as a dependent variable to analyze the logical regression model. Since the dependent variable in the first question is only momentum, and the dependent variable here is the performance fluctuation of momentum, we have made certain modifications and re-assigned independent variables. The new assignment table is shown in the figure below.

Table 10 New grading situation

Object	Indicators	Description	New grading situation
Serve Situation	AC	P1_Ace change	P1_Ace +30/P2_Ace -15
	DFC	P1_Double faault change	P1_Double fault -30/P2_Douuble fault +15
	SNC	P1_Serve_no change	Sever=1:Serve_no：1 +5/：2 -5
Stroke situation	NPC	P1_net_pt_won change	P1_net_pt_won +10/P2_net_pt_won -10
	WC	P1_Winner change	P1_Winner +12/P2_ Winner -12
	UEC	P1_unf_error change	P1_unf_error -8/P2_unf_error +8
Win situation	SC	P1_Score_change	P1_Score:30-40 40-40 AD-40 +40/40-30 40-40 40-AD -40
	BUC	P1_break_pt_change	P1_Break_pt_won +60/miss -20 P2_Break_pt_won -20/miss +60

The following is to build a specific model:

For binary classification problems, given that the values of $w^T x + b$ are continuous, it cannot fit discrete variables. We can consider using it to fit conditional probabilities $p(Y = 1|x)$, since probability values are also continuous.

The ideal is the unit step function:

$$p(y = 1|x) = \begin{cases} 0, & z < 0 \\ 0.5, & z = 0 \\ 1, & z > 0 \end{cases} \tag{19}$$

But this step function is not differentiable, and the logarithmic probability function is a common alternative:

$$y = \frac{1}{1+e^{-(w^T x+b)}} \tag{20}$$

The above formula can be formed

$$ln\frac{y}{1-y} = w^T x + b \tag{21}$$

Where y is the probability that x is a positive example, and 1-y is the probability that x is its counterexample. The ratio of the two is called odds. Therefore, in fact, the dependent variable value in logistic regression should be odds.

Treat y as a posterior-like probability estimate and rewrite the formula as follows:

$$w^T x + b = \ln\frac{P(Y=1|x)}{1-P(Y=1|x)} \tag{22}$$

$$P(Y = 1|x) = \frac{1}{1+e^{-(w^T x+b)}} \tag{23}$$

For multi-classification logistic regression, we uses the comparison of a certain class with the remaining classes as a binary classification problem by default. N classes are classified n-1 times, n-1 binary classification models are obtained, and the probability corresponding to each binary classification is obtained. The class with the highest probability is used as the prediction result of the new sample.

We select Sinner's four match data for logistic regression model analysis, and observe the momentum performance fluctuation by whether momentum increases or decreases.

Based on the analysis of the model, we draw two tables and draw some conclusions respectively, and generate the most concerned regression coefficient.

Table 11 Binary logistic regression results

Test group =-1.0	Regression coefficient	Standard error	Wald	P	OR	OR value 95% confidence interval

						Upper limit	lower limits
constant	0.188	0.089	4.466	0.035**	1.207	1.014	1.436
p1_ace_change	-0.115	0.045	6.519	0.011**	0.891	0.816	0.974
p1_double	0.015	0.019	0.624	0.430	1.015	0.978	1.053
p1_serve_no_change	-0.112	0.022	25.478	0.000**	0.894	0.856	0.934
p1_net_pt_change	-0.215	0.048	20.227	0.000**	0.807	0.735	0.886
p1_winner_change	-0.124	0.018	45.49	0.000**	0.883	0.852	0.916
p1_unf_error_change	-0.063	0.02	10.145	0.001**	0.939	0.904	0.976
p1_soore_change	0.01	0.016	0.403	0.525	1.01	0.979	1.043
p1_break_up_change	-0.005	0.005	0.96%	0.326	0.995	0.984	1.005

Dependent variable: player1_momentum_performance_fluctuation_determine_categorization

Note: ***, ** and * represent significance levels of 1%, 5% and 10% respectively

The analysis of this statistical model shows that in tennis matches, serve score rate, net point change, win point change and unforced error change are negatively correlated with the fluctuation of player performance momentum, that is, the improvement of these indicators is often associated with an increase in the stability of player performance. Variables with an odds ratio (OR) of less than 1 show that an increase per unit (actually an increase in performance) leads to an increase in the likelihood that a player's performance will fluctuate less. These results have practical implications for understanding performance factors in matches and how they affect player momentum.

Table 12

Accuracy rate	Recall rate	Precision rate	F1	AUC
0.739	0.739	0.74	0.739	0.815

In the analytical context of a tennis match, the consistency of the model meant that it was able to identify successful rounds or movements in a balanced way, but not with very high accuracy. AUC values indicate that the model is better at distinguishing between positive and negative outcomes in a game, such as points versus turnovers. However, since none of the indicators are close to perfect, the models may need to be further tweaked to more accurately predict match outcomes or assess player performance.

So far, we have used the logistic regression model to get the specific relationship and expression of the performance fluctuation of the independent variables (serving, hitting, winning) and the dependent variable momentum.

Therefore, we can definitely predict the performance fluctuation of players' momentum change in tennis matches by using logistic regression model, but the robustness of this model needs further study.

7.2 CRITIC weight method

In order to study which indicators of independent variables have stronger correlation, we apply the CRITIC weight method.

(1) Indicator variability

Expressed as standard deviation, Sj represents the standard deviation of the JTH indicator.

$$\begin{cases} \overline{x}_j = \frac{1}{n}\sum_{i=1}^{n} x_{ij} \\ S_j = \sqrt{\frac{\sum_{i=1}^{n}(x_{ij}-\bar{x}_j)^2}{n-1}} \end{cases} \tag{24}$$

In the CRITIC method, standard deviation is used to represent the difference fluctuation of the internal value of each indicator. The larger the standard deviation is, the greater the numerical difference of the indicator, the more information can be projected, and the stronger the evaluation intensity of the indicator itself, so more weight should be assigned to the indicator.

(2) Index conflict

Expressed in the form of correlation coefficient, rij represents the correlation coefficient between evaluation indicators i and j.

$$R_j = \sum_{i=1}^{p}(1 - r_{ij}) \tag{25}$$

The correlation coefficient is used to represent the correlation among indicators. The stronger the correlation is with other indicators, the smaller the conflict between the indicator and other indicators. The more the same information is reflected, the more repeated the evaluation content can be reflected, and to some extent, the evaluation intensity of the indicator is weakened, and the weight assigned to the indicator should be reduced.

(3) Information volume

The larger the Cj is, the greater the role of the JTH evaluation index in the whole evaluation index system, so more weight should be assigned to it.

$$C_j = \sum_{i-1}^{p}(1 - r_{ij}) = S_j \times R_j \tag{26}$$

(4) Weight

So the objective weight of the JTH index Wj

$$W_j = \frac{C_j}{\sum_{j=1}^{p} C_j} \tag{27}$$

Through the weight analysis of CRITIC, we get the following table and pie chart:

Table 13 CRITIC weight table

item	Index variability	Index conflict	Amount of information	Weight (%)
p1_ace_change	7.671	6.207	47.614	15.322
p1_double_fault_change	4.528	6.329	28.66	9.223
p1_serve_no_change	3.875	6.233	24.15	7.771
p1_net_pt_change	3.393	6.595	22.38	7.202
p1_winner_change	7.068	5.912	41.784	13.446
p1_unf _error_change	4.164	6.483	26.996	8.687
p1_score_change	5.919	6.681	39.546	12.726
p1_break_up_change	11.606	6.861	79.628	25.624

From the data in the table above, we can find that the weights of break_up_change, ace_change, winner_change, and score_change are relatively high, indicating that these factors are more correlated

7.3 Competition Recommendations

We have found the relationship between real-time fluctuations in momentum and multiple indicators in the first question, so we suggest that when a player plays a new match with another player, the following factors should be considered:

Compare opponents: Analyze similar data about the opponent you will be playing to see if there are similar patterns of momentum shifts or vulnerabilities in certain areas.

Intensive training: Based on the results of the analysis, strengthen the player's performance at possible turning points, such as improving the quality of the key points or strengthening the mental preparation.

Adjust the game strategy: According to the opponent's vulnerability and their strengths, adjust the game strategy, such as increasing the attack when the opponent's physical decline or psychological instability.

Fore-and-backhand and main hand: Each player's main hand is different, and the ability of the forward and reverse strokes is also different, and the vulnerable direction of the opponent can be targeted according to the above information.

In response to the analysis of Sinna in the first question, we can consider that when facing Sinna as an opponent, we need to watch his game video more before the match, observe his hitting points, and analyze the characteristics of his forehand and backhand. We found that the forehand strength is more

even, the forehand strength is stronger, but the backhand can often hit high-quality winning points, we can selectively mobilize, hit repeated drops. At the same time, we can observe that Sinna's ability in the bottom line and in front of the net is very outstanding, especially in the bottom line of the holding ability and the ability to mobilize. Against such a strong opponent, adjusting the mentality to put pressure on the other side at the crucial turning point is the key to winning.

8 Model Evaluation

According to logistic regression model,

$$p = \frac{e^{\beta_0+\beta_1 x_1+\beta_2 x_2+\ldots+\beta_m x_m}}{1+e^{\beta_0+\beta_1 x_1+\beta_2 x_2+\ldots+\beta_m x_m}} \tag{28}$$

We re-selected Daniil Medvedev's data to test the logistic regression model.

By comparing the predicted momentum performance fluctuation with the actual momentum performance fluctuation and calculating the average value of the coincidence degree of the four match data, the result is roughly 70.99%.

8.1 Sensitivity Analysis

When the model is not accurate enough to fit the new match and the performance is not good, we should include the ATP men's singles world ranking in the list, because the psychological pressure is bound to be different when facing opponents with different rankings, which will lead to different momentum at the beginning and the change of momentum during the process. The real-time ranking we found is shown in the following table: (NA indicates that the player has retired) diagram We can see through the real-time ranking of the Wimbledon 32 strength gap is large, the world's top ten are basically in them, but there are also several players ranked in the 100, and even the qualifiers play to the match. When a player seeded in the top 32 of the world comes up against a player much lower in the rankings, the mental side becomes more dominant and the initial momentum increases accordingly. However, if the seeded player is behind during the match, the momentum will fluctuate more significantly, because it is obviously not in line with their psychological expectations. From this we can conclude that the ATP world ranking can be factored into future models.

The real-time ranking we found is shown in the following table: (NA indicates that the player has retired)

We have found the relationship between real-time fluctuations in momentum and multiple indicators in the first question, so we suggest that when a player plays a new game against another player, he should compare the opponent, intensify training,

Adjust the game strategy, observe the opponent's forehand and backhand and main hand. In response to the analysis of Sinna in the first question, we can consider that when facing Sinna as an opponent, we need to watch his game video more before the match, observe his hitting points, and analyze the characteristics of his forehand and backhand. We found that the forehand strength is more even, the forehand strength is stronger, but the backhand can often hit high-quality winning points, we can selectively mobilize, hit repeated drops. At the same time, we can observe that Sinna's ability in the bottom line and in front of the net is very outstanding, especially in the bottom line of the

holding ability and the ability to mobilize. Against such a strong opponent, adjusting the mentality to put pressure on the other side at the crucial turning point is the key to winning.

Player	World Ranking
Carlos Alcaraz	2
Novak Djokovic	1
Daniil Medvedev	3
Stefanos Tsitsipas	10
Holger Rune	7
Andrey Rublev	5
Jannik Sinner	4
Alexander Zverev	6
Frances Tiafoe	14
Hubert Hurcakz	8
Tommy Paul	15
Grigor Dimitrov	13
Alejandro Davidovich Fokina	23
Jiri Lehecka	31
Christopher Eubanks	32
Laslo Djere	34
Daniel Rlahi Galan	79
Guido Pella	NA
Denis Shapovalov	137
Alexander Bublik	27
Lorenzo Musetti	26
Stan Wawrinka	56
Roman Safiullin	36
Nicolas Jarry	20
Matteo Berrettini	124
Marton Fucsovics	82
Christopher O'Connell	64
Quentin Halys	100
Mikael Ymer	NA
Liam Broady	97
David Goffin	134
Maximilian Marterer	92

Figure 13 World ranking table

8.2 Robustness Analysis

Since the overall rules of women's tennis match are basically the same as men's tennis match, except for 2 wins in 3 sets in Grand Slam tournaments, the model we developed is generally applicable to women's tennis match to a large extent.

Since the Grand Slam has its own special rules, men's singles matches are won 3 out of 5 sets and 10 out of the tiebreak in the deciding set. In addition to the tennis Grand Slam, the rules of men's singles match and women's singles match are the same, which is consistent with 1). Therefore, the model we developed is generally applicable to all major championships to a large extent.

Tennis is played on hard courts, with the Australian Open and US Open among the four Grand Slams; Grass courts, the Wimbledon tennis championships of the four Grand Slams; And clay courts, the French Open of the four Grand Slams. But the rules are the same in different courts, but the speed of the ball will be slightly different on different courts, the speed of the hard court will be faster, the clay court will be slower, and there will be ball marks. In other words, the adaptability of each player to each field is also different, which will affect the model results more or less. But in general, our model has a strong universality to different tennis courts.

The rules of other sports such as table tennis are quite different from those of tennis. For example, table tennis wins 4 out of 7 sets with 11 points per set, while tennis is 2 out of 3 sets or 3 out of 5 sets with a maximum of 13 sets per set and at least 4 points per set. Tennis and table tennis are also very different in terms of the size of the court, the size of the

racket and the distance of the run, and tennis has a special rule of "second serve". Therefore, the model we developed is less applicable to other sports.

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9 Conclusion

This paper studies momentum with tennis ball as the carrier. momentum is used as the core throughout the whole paper. momentum is used as a bridge in the first two questions, and its dependent variable in the first question is transformed into an independent variable in the second question, which is analyzed by Topsis entropy weight method and spearman correlation analysis. In the third question, the momentum fluctuation is re-regarded as the dependent variable, and further explained by logistic regression and critic entropy weight method. Finally, the fourth question backtests the model used in the third question and introduces new indicators for momentum.

10 Memorandum

To: Traders

From: MCM Team 2405303

Subject: The Momentum Mystery: Demystifying the mystery

Date: February 6, 2024

Yours sincerely,
Team # 2405303

Grand Slam

Wimbledon Championships

It is held annually in June or July in Wimbledon, a suburb of London, UK

US.Open

Held annually in New York, USA from the end of August to the beginning of September

French Open

Held annually from May to June at the Roland Garros Stadium in Paris, France

Australian Open

Held annually in the last two weeks of January at Melbourne Sports Park in Victoria, Australia

Wimbledon Championships, abbreviated as "Wimbledon".

Association in 1877, it is held in Wimbledon, a suburb of London, and is also the only grass competition among the Four Slams. The Wimbledon Tennis Championships are usually held in June or July, and the eligibility for participation is determined bas

French Open, abbreviated as "French Open".

It is a tennis Grand Slam event held at the Roland Garros Stadium in Paris, France. Usually held from May to June each year, it is the second Grand Slam tournament to be held each year.

Australian Open, abbreviated as "Australian Open".

The competition is usually held in the last two weeks of January each year at Melbourne Sports Park in Victoria, Australia. It is the first of the four Grand Slams held each year and one of the two most prestigious tennis hard court events.

U. S. Open, abbreviated as "US Open".

It is the fourth and final tennis Grand Slam event of the year, usually held from the end of August to early September at Flushing Park in New York, USA.

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